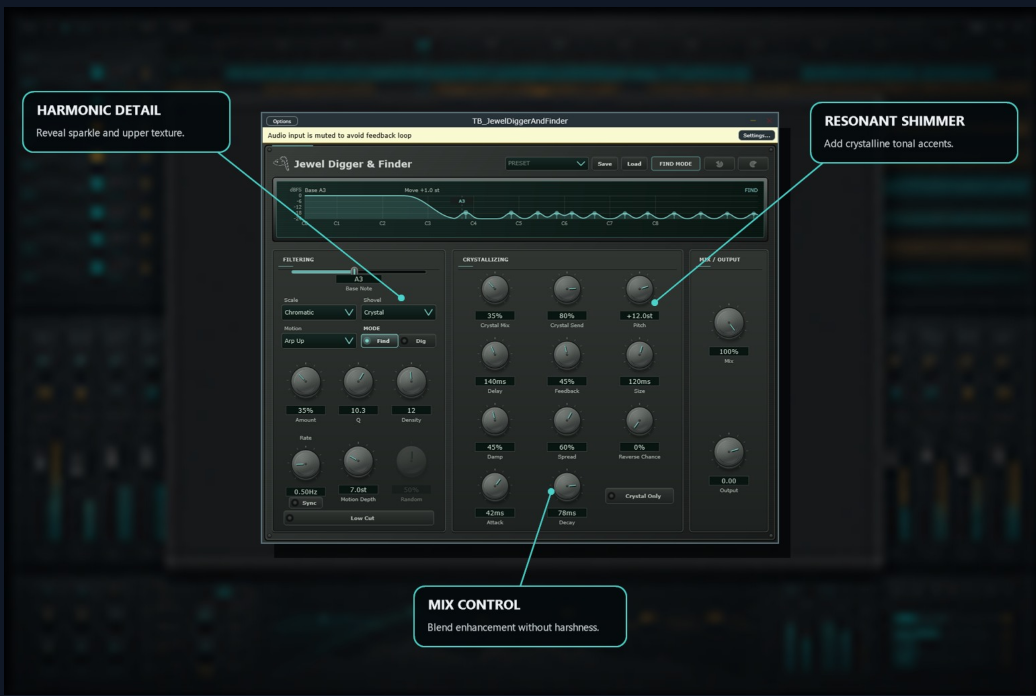
TB Jewel Digger & Finder

DETAILED ENGLISH USER MANUAL

A scale-aware resonant filter-motion processor with Find/Dig modes and a parallel granular crystal engine.



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Host-visible endpoints documented: 30

1. Product purpose

A scale-aware resonant filter-motion processor with Find/Dig modes and a parallel granular crystal engine.

Broad shimmer reverbs often brighten everything and can obscure the harmonic center. TB Jewel Digger & Finder selects musically related bands above a base note, emphasizes or removes them, and sends that chosen material into a controllable crystal layer.

What result it creates

- Harmonic emphasis or subtractive comb motion
- Scale-aligned resonance patterns
- Tempo-synced or free filter animation
- Pitch-shifted, delayed, reversed crystal grains

Signal flow

Input -> scale target generator -> Find or Dig filter bank -> crystal send and grain engine -> Crystal Only option -> final Mix and Output

2. Complete feature guide

Base Note and Scale

Base Note selects C0-C8 and Scale limits target pitch classes. Set them to the song center or deliberately choose a contrasting map.

Find and Dig

Find emphasizes selected bands; Dig subtracts them to create moving notches and comb textures. Base Low Cut determines whether the area below the base note is protected or filtered.

Shovel, Density, Q, Amount

Shovel chooses the weighting/detune character. Density controls target count, Q controls width, and Amount sets filter intensity.

Motion system

Still, arpeggios, ramps, sine, square, triangle, musical random, and chaos move the targets. Depth controls travel; Rate can run freely or Sync to host divisions.

Crystal engine

Crystal Send feeds selected material to grains. Pitch, Grain Size, Attack, Decay, Delay, Feedback, Spread, Damp, Random, and Reverse Chance define the layer.

Crystal Mix and Crystal Only

Crystal Mix controls crystal return level. Crystal Only removes the normal filtered wet path but does not remove dry signal when the global Mix is below 100 percent.

Output workflow

Use Mix for dry/final-wet balance and Output for final trim. Programs range from harmonic lanterns to cathedral shimmer and octave rain.

3. Quick start

1. Set Base Note and Scale.

2. Choose Find or Dig.

3. Set Shovel, Density, Q, and Amount.

4. Select Motion and tune Depth/Rate or enable Sync.

5. Raise Crystal Send and Crystal Mix.

6. Set Pitch and grain envelope controls.

7. Use Crystal Only if only the granular return is required, then finish with Mix and Output.

4. Practical workflows

Harmonic shimmer

1. Choose Find.

2. Match Base Note and Scale to the song.

3. Use moderate Density and Q.

4. Set Crystal Pitch to +12 semitones and raise Crystal Mix gradually.

Expected result: A luminous octave layer grows from harmonically selected content.

Moving spectral carve

1. Choose Dig.

2. Select an arpeggio or ramp Motion.

3. Raise Amount and Q.

4. Keep Crystal Mix low or off.

Expected result: The source develops animated scale-related notches without a large reverb tail.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Wrong key or scale choices can create intentional but dissonant resonances.
- High feedback and dense grains can accumulate level.
- Tempo sync requires valid host tempo information.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 30 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Amount VST3 ID 1659428187	0.000 to 1.000	0.350	Automatable	Sets how strongly the named process affects the signal.
Base Low Cut VST3 ID 1124829311	Off, On	Off	Automatable	Reduces low-frequency content in the associated audio or detector path.
Base Note VST3 ID 250297620	C0 to C8	A3	Automatable	Sets the principal frequency, note, or spectral center used by this function.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Crystal Damp VST3 ID 1066425142	0.000 to 1.000	0.450	Automatable	Reduces high-frequency content or shortens high-frequency decay in the associated path.
Crystal Mix VST3 ID 657872710	0.000 to 1.000	0.350	Automatable	Sets the balance between the original signal and the complete processed path.
Crystal Only VST3 ID 1066765314	Off, On	Off	Automatable	Host-automatable control for Crystal Only. Its full range and default are shown in this table; use it with the related feature section above.
Crystal Send VST3 ID 517603273	0.000 to 1.000	0.800	Automatable	Sets gain for the named input, output, band, or parallel send/return path.
Crystal Spread VST3 ID 928292937	0.000 to 1.000	0.600	Automatable	Sets stereo spread or the lateral distribution of the processed signal.
Density VST3 ID 1552717032	1 to 24	12	Automatable	Controls granular density, dispersion, or random variation for the named process.
Division VST3 ID 564265389	1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1/1	1/4	Automatable	Host-automatable control for Division. Its full range and default are shown in this table; use it with the related feature section above.
Filter VST3 ID 594214203	Find, Dig	Find	Automatable	Host-automatable control for Filter. Its full range and default are shown in this table; use it with the related feature section above.
Grain Attack VST3 ID 712184483	0.010 to 0.750	0.350	Automatable	Sets how quickly the associated detector, envelope, grain, or dynamics stage responds at onset.
Grain Decay VST3 ID 510196735	0.010 to 0.950	0.650	Automatable	Sets how long the associated detector, envelope, reverb, or resonant process takes to recover or decay.
Grain Delay VST3 ID 510205384	0.00 to 1000.00	140.00	Automatable	Controls granular density, dispersion, or random variation for the named process.
Grain Feedback VST3 ID 779248928	0.0% to 100%	45%	Automatable	Returns processed signal to its own input, extending or intensifying the effect.
Grain Size VST3 ID 778919708	15.00 to 400.00	120.00	Automatable	Shapes spatial density, resonant spread, or diffuse body for the named process.
Mix VST3 ID 108124	0.000 to 1.000	1.000	Automatable	Sets the balance between the original signal and the complete processed path.
Motion VST3 ID 1079164854	Still, Arp Up, Arp Down, Ramp Up, Ramp Down, Sine, Square, Triangle, Musical Random, Chaos	Arp Up	Automatable	Host-automatable control for Motion. Its full range and default are shown in this table; use it with the related feature section above.
Motion Depth VST3 ID 1422555072	0.00 to 24.00	7.00	Automatable	Sets how strongly the named process affects the signal.
Output VST3 ID 1141971201	-18.00 to 6.00	0.00	Automatable	Sets or limits the final output level after the plug-in's main processing.
Pitch VST3 ID 521414533	-24.00 to 24.00	12.00	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.
Program VST3 ID 1886548852	Initialize, Harmonic Lantern, Frozen Constellation, Amber Drill, Reverse Shards, Phase Dust, Cathedral Shimmer, Octave Rain, Fifth Halo	Initialize	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.

Parameter	Range / options	Default	Host status	Function
Q VST3 ID 113	0.35 to 24.00	6.00	Automatable	Sets bandwidth or resonance sharpness for the named filter or analysis band.
Random VST3 ID 1071006459	0.000 to 1.000	0.500	Automatable	Controls granular density, dispersion, or random variation for the named process.
Rate VST3 ID 3493088	0.020 to 64.000	0.500	Automatable	Sets the speed of modulation, movement, or repeated change.
Rate Sync VST3 ID 422275995	Off, On	Off	Automatable	Sets the speed of modulation, movement, or repeated change.
Reverse Chance VST3 ID 1099846370	0.00 to 100.00	0.00	Automatable	Sets whether or how often audio fragments are played in reverse.
Scale VST3 ID 109250890	Chromatic, Major, Minor, Major 7, Minor 7, Diminished, Augmented, Pentatonic Major, Pentatonic Minor, Mixolydian	Chromatic	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Shovel VST3 ID 1244338339	Crystal, Hollow, Bow, Hammer, Metal, Dirt	Crystal	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.